

Suitable models generate rich families in every Banach space

Literature answers to Questions 2 and 3 of arXiv:1308.1818

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Status: literature-already-answered, full positive answers to Questions 2 and 3. Cúth's later Theorems 4 and 15 remove all structural restrictions: suitable models generate nice rich families in every Banach space, and the suitable-model and rich-family formulations of separable determination refine one another.

The source questions

Let $X_M = \overline{X \cap M}$. The source says that *suitable models generate nice rich families in X* if, for every prescribed finite list of formulas and countable parameter set, one can choose suitable models M so that

$$\{X_M : M \in \mathcal{M}\}$$

is a cofinal, σ -closed family of separable subspaces and $M \subset N$ exactly when $X_M \subset X_N$.

Question 2 on source PDF page 7 asks whether this holds for every Banach space, specifically including $\ell_\infty^c(\Gamma)$. Question 3 on source PDF page 10 asks whether separable determination by suitable models and by rich families is equivalent without assuming a fundamental minimal system or density $\aleph_1$ [1].

The explicit later answer

The abstract of [2] announces equivalence in Banach spaces and positive answers to two questions of Cúth–Kalenda. The exact results are:

- Theorem 15 (supporting PDF page 5): if a topological space is homeomorphic to a Banach space, then suitable models generate nice rich families in it. Taking the space to be any Banach space proves source Question 2, including the stated $\ell_\infty^c(\Gamma)$ case.
- Theorem 4 (supporting PDF page 2): for a Banach space and a family $\mathcal{F}$ of closed separable subspaces, every rich $\mathcal{F}$ contains a suitable-model-large subfamily, and every suitable-model-large $\mathcal{F}$ contains a rich subfamily. Applied to any family on which a statement is separably determined, these refinements preserve the statement and prove source Question 3.

The supporting paper itself explicitly identifies Theorem 15 as a positive answer to source Question 2 and Theorem 4 as yielding the positive answer to source Question 3.

Proof intuition

Hilbert spaces already satisfy the desired model/subspace order correspondence. Toruńczyk's topological classification supplies, for an arbitrary infinite-dimensional Banach space X , a homeomorphism $f : X \rightarrow H$ onto a Hilbert space of the same density. Choose the models to contain f and f^{-1} . Elementarity then gives

$$f(X \cap M) = H \cap M, \quad f(X_M) = H_M.$$

Consequently cofinality, closure under increasing countable unions, and the equivalence $M \subset N \iff H_M \subset H_N$ transfer through f to X . This proves Theorem 15. Theorem 4 then converts either large-family formulation into the other by passing to a subfamily, so every separable-determination equivalence survives unchanged.

Scope boundary

The source also asks whether the particular spanning system used as a sufficient condition in its Lemma 2.6 is *necessary* (Question 1). The later theorem bypasses that condition by a topological conjugacy; it does not construct such a spanning system or prove its necessity. Thus this packet classifies only Questions 2 and 3 as fully answered.

References

- [1] M. Cúth and O. F. K. Kalenda, *Rich families and elementary submodels*, arXiv:1308.1818v3 (2013), Questions 2 and 3.
- [2] M. Cúth, *Separable determination in Banach spaces*, arXiv:1608.03685v2 (2016), Theorems 4 and 15.